

Oja-Type Plasticity and Neural Signal Reconstruction in Recurrent Networks

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1 INTRODUCTION

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3 BACKGROUND

3.1 FIXED-WEIGHT RECURRENT NETWORKS

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3.4 GIRKO'S CIRCULAR LAW

Fixed-weight recurrent networks are typically studied through deterministic methods such as fixed-point stability analysis. However, with Oja-type plasticity, we must study a higher-dimensional augmented system of weights and neural activity with complex interactions. Probabilistic methods allow us to relax some of the dependence on exact trajectory-level analysis and instead study typical behavior across ensembles of networks. Motivated by the use of random weight initializations in reservoir computing, we utilize Girko's circular law to analyze the evolution of the eigenvalues of our weight matrix.

Let $W^{(n)} = (W_{ij}) \in \mathbb{R}^{n \times n}$ be a random matrix whose entries are independent and identically distributed (i.i.d.) real random variables satisfying

$$\mathbb{E}[W_{ij}] = 0, \quad \mathbb{E}[W_{ij}^2] = \frac{g^2}{n}$$

Let $\lambda_1, \dots, \lambda_n \in \mathbb{C}$ denote the eigenvalues of $W^{(n)}$, and define the empirical spectral measure

$$\mu_n = \frac{1}{n} \sum_{k=1}^n \delta_{\lambda_k}.$$

Then Girko's circular law states that, as $n \rightarrow \infty$,

$$\mu_n \Rightarrow \mu_{\text{circ}},$$

where $\Rightarrow$ denotes weak convergence of probability measures and

$$d\mu_{\text{circ}}(z) = \frac{1}{\pi g^2} \mathbf{1}_{\{|z| \leq g\}} d^2 z$$

is the uniform probability measure on the disk of radius g in $\mathbb{C}$.

For a sufficiently large system of n neurons with weights sampled from $\mathcal{N}(0, \frac{g^2}{n})$, Girko's Circular Law predicts that the eigenvalues of the neural weight matrix distribute uniformly on the disk

$$D_g = \{z \in \mathbb{C} \mid |z| \leq g\}.$$

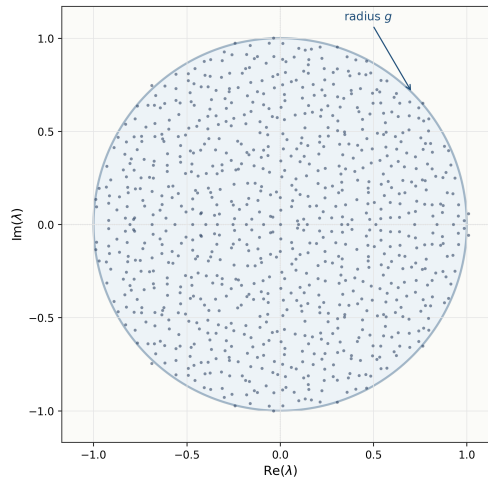


Figure 3.1: Eigenvalues of a random weight matrix $W \in \mathbb{R}^{n \times n}$, $W_{ij} \sim \mathcal{N}(0, \frac{g^2}{n^2})$, $n = 900$ uniformly distributed on the disk D_g of radius g in accordance with Girko's circular law.

The "gain" parameter, g , thus controls spectral radius $\rho(W) \approx g$ which in turn determines strength of recurrent feedback.

4 THEORETICAL ANALYSIS

4.1 FIXED-POINT ANALYSIS TODO: FIX TITLE

Suppose that we randomly initialize the interneural weights for a neural network as above. Formally, encode this as $W \in \mathbb{R}^{n \times n}$ with $W_{ij} \sim \mathcal{N}(0, \frac{g^2}{n})$ where n is the number of neurons in our network. To simplify analysis, a neuron is also initialized with a random self-recurrent connection. Throughtout our analysis in this paper, we shall use $\tanh$ as our activation function for its nonlinear, smooth, zero-centered, and bounded behavior. First, we briefly analyze the undriven behavior of the fixed-weight system, that is, we analyze

$$\tau \dot{\mathbf{x}} = -\mathbf{x} + \tanh(W\mathbf{x}).$$

Here, we apply hyperbolic tangent elementwise; that is,

$$\tau \dot{x}_i = -x_i + \tanh(\sum_j W_{ij} x_j)$$

Linearizing around the origin yields the Jacobian

$$J(0) = \frac{1}{\tau}(-\text{Id} + W).$$

Each eigenvalue λ of W corresponds to eigenvalue

$$\mu = \frac{\lambda - 1}{\tau}$$

of $J(0)$. By Girko's circular law, the eigenvalues of W approximately fill the disk $|\lambda| \leq g$, so the Jacobian eigenvalues fill the shifted disk $|\tau\mu + 1| \leq g$. The rightmost point of this disk lies at

$$\Re(\mu)_{\max} \approx \frac{g-1}{\mu}$$

Consequentially, we expect that the origin is stable for $g < 1$ and unstable for $g > 1$, with $g = 1$ some Jacobian eigenvalues cross the imaginary axis and the origin becomes unstable; perturbations can be amplified by recurrent feedback. Thus, $g = 1$ here serves as a bifurcation point.

4.2 STABILITY AND EXISTENCE ARGUMENTS

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4.3 AVERAGING ANALYSIS AND SLOW-MANIFOLD DYNAMICS

A key assumption in our analysis is the separation of time scales in neural activity and Oja-type learning. Assume that

$$\tau \ll \frac{1}{\eta}$$

so that the neural activity evolves on a much faster time scale than that of the weights. In our simulations, we use $\eta = 0.01$ ($1/\eta = 100$) and $\tau = 1$.

On the fast activity timescale, W is approximately constant. Section 4.2 then predicts that $\mathbf{x}(t)$ settles into a unique periodic orbit, $\mathbf{x}(t; W)$, of frequency ω determined by the current weight matrix W . Now, we shall use averaging theory to characterize dynamics on the slow manifold. Averaging over one drive period $T_d = 2\pi/\omega$ yields the slow-manifold dynamics

$$\dot{W}_{ij} \approx \eta[C_{ij}(W) - C_{ii}(W)W_{ij}] - \lambda W_{ij}$$

where

$$C(W) = \frac{1}{T_d} \int_0^{T_d} \mathbf{x}(t; W) \mathbf{x}(t; W)^T dt$$

is the time-averaged correlation matrix of the neural activity. Setting $\dot{W}_{ij} = 0$ yields the fixed point condition

$$W_{ij}^* = \frac{\eta C_{ij}(W^*)}{\eta C_{ii}(W^*) + \lambda}.$$

For $\mathbf{x}(t) = \mathbf{x}$, that is constant neural activity, the above equation reduces to

$$W_{ij}^* = \frac{x_i x_j}{\eta x_i^2 + \lambda} = u_i x_j, \quad u_i = \frac{\eta x_i}{\eta x_i^2 + \lambda},$$

so $W = u x^T$. Thus, we have that a fixed point in this neural-weight augmented system corresponds to a rank-one weight matrix.

For sinusoidal drive at frequency ω , we have that every **explain before why frequency is omega** neuron responds at the same frequency $x_i(t) \approx r_i \sin(\omega t + \theta_i)$. We then have that the time-averaged correlation matrix is

$$C_{ij}(W) \approx \frac{1}{2} r_i r_j \cos(\theta_i - \theta_j)$$

Letting $a_i := r_i \cos(\theta_i)$ and $b_i := r_i \sin(\theta_i)$, we have

$$C(W) \approx \frac{1}{2} (a a^T + b b^T)$$

and is thus of rank at most 2. Now, when in the saturated regime of weight dynamics, $x_i \rightarrow \pm 1$ is a fixed point **CHARACTERIZE STABILITY OF THIS FIXED POINT AND FORMALLY JUSTIFY**. Thus, $C_{ii} = x_i^2 \rightarrow 1$ as our activity saturates. We thus have the following

$$W^* \approx \frac{\eta}{\eta + \lambda} C(W^*)$$

and since $C(W)$ is positive semi-definite,

$$\rho(C(W)) \leq \max_i \|\lambda_i\| \leq \text{tr}(C) \approx n$$

under saturation, and so

$$\rho(W^*) \leq \frac{\eta}{\eta + \lambda} \text{tr}(C) \approx \frac{n\eta}{\eta + \lambda}.$$

So, we have approximated an upper bound on the spectral radius of the saturated system.

Adi can you check this If we additionally assume that under saturation, the phases of the neural activities approximately align, then $\cos(\theta_i - \theta_j) \approx 1$ and so

$$C(W^*) \approx \frac{1}{2}(aa^T + bb^T) \approx \frac{1}{2}rr^T$$

collapses to a rank 1 matrix. We obtain an explicit estimate of spectral radius

$$\rho(W^*) \approx \frac{\eta}{\eta + 2\lambda} \|r\|^2$$

under this approximation. Since each neuron saturates to $r_i^2 \approx x_i^2 = 1$, we obtain

$$\rho(W^*) \approx \frac{n\eta}{\eta + 2\lambda}$$

If we use the parameters from our numerical simulations, specifically $n = 50$, $\eta = 0.01$, and $\lambda = 0.001$, the first approximation yields

$$\rho(W^*) \leq 45.5$$

and the second approximation yields

$$\rho(W^*) \approx 41.7.$$

4.4 EFFECTIVE RANDOM-MATRIX GAIN

Although the nominal spectral radius $\rho(W)$ characterizes the initial random matrix, the stability of the driven system is governed by $\rho(D(t)W)$, the spectral radius of the effective recurrent matrix $D(t)W$. Because the entries of $D(t)W$ are $d_i(t)W_{ij}$, and W_{ij} has variance g^2/n , conditional on $D(t)$ we have that

$$\text{Var}((D(t)W)_{ij}) = \frac{g^2 d_i(t)^2}{n}. \quad (4.1)$$

Applying Girko's law to $D(t)W$ motivates what we denote *effective gain*

$$g_{\text{eff}}(t) \approx \frac{\|W(t)\|_F}{\sqrt{n}} \sqrt{\frac{1}{n} \sum_{i=1}^n d_i(t)^2}. \quad (4.2)$$

Where we approximate

$$\mathbb{E}[\|W(t)\|_F^2] = \sum_{i,j} \mathbb{E}[W_{ij}^2(t)] = n^2 \cdot \frac{g^2}{n} = ng^2$$

under the assumption that W_{ij} remain approximately i.i.d of mean 0 and variance g^2/n . Because W is not stationary, this approximation becomes weaker as the system evolves; however, numerical simulations demonstrate the effectiveness of g_{eff} in approximation $\rho(D(t)W)$ in the short-term. When neurons are unsaturated ($d_i \approx 1$), $g_{\text{eff}} \approx g$. Under strong periodic drive, neurons saturate ($|h_i| \rightarrow \infty$, $d_i \rightarrow 0$), pulling g_{eff} below g regardless of spectral radius of W .

5 NUMERICAL SIMULATIONS

May need to fix titles for the graphs This section provides a series of numerical results which test a variety of hypotheses and properties of the system with Oja-type plasticity. All simulations use networks of $n = 50$ neurons with $\tau = 1$, $\eta = 0.01$, $\lambda = 0.001$, sinusoidal drive amplitude $A = 1$, and frequency $\omega = 1$ unless otherwise specified. Random phase offsets $\phi_i \sim \text{Uniform}(0, 2\pi)$ are drawn independently for each neuron so that the population-averaged input correlation matrix is not trivially rank-one by construction. Results are averaged over 5 random seeds with error bars showing one standard deviation.

5.1 OJA LEARNING DOES NOT SELF-ORGANIZE TO THE EDGE OF CHAOS

A motivation for studying recurrent neural networks with Oja-type plasticity was to study the hypothesis that perhaps such a learning rule may reorganize weights to the edge of chaos, allowing for optimal signal reconstruction. To test this hypothesis, we sweep initial spectral radii $g_0 \in \{0.3, 0.5, 0.7, 1.0, 1.5, 2.0, 2.5\}$ and run the full coupled system

$$\begin{aligned} \tau \dot{x}_i &= -x_i + \tanh\left(\sum_j W_{ij} x_j + A \sin(\omega t + \phi_i)\right), \\ \dot{W}_{ij} &= \eta x_i (x_j - x_i W_{ij}) - \lambda W_{ij}, \end{aligned} \quad (5.1)$$

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for $T = 400$ time units under both sinusoidal drive ($A = 1$) and autonomous dynamics ($A = 0$).

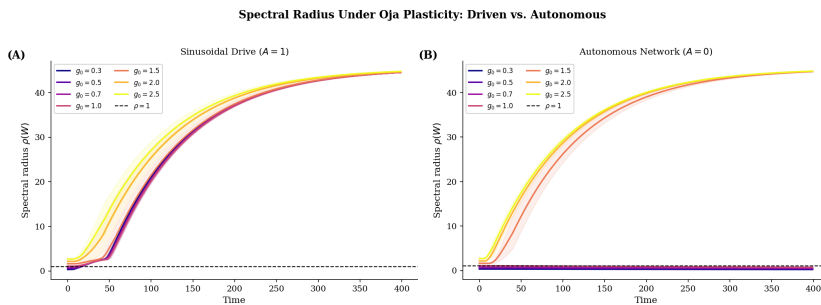


Figure 5.1: Spectral radius $\rho(W(t))$ over time for all initial conditions g_0 under sinusoidal drive (left) and autonomous dynamics (right).

Fix image formatting and text captions Under sinusoidal drive ($A = 1$), the spectral radius of the blows up for all initial conditions ($\rho(W) \rightarrow 45$ for every initial g_0), with all trajectories converging within $T \approx 150$ time units (Figure 5.1). The drive erases all memory of the initial condition. The autonomous system ($A = 0$), on the other hand, exhibits a bifurcation at $g_0 = 1$.

These results suggest that addition of periodic input erases this bifurcation, removing the stable branch near $\rho(W) = 1$. The mechanism reflects the analysis in Section 4.3 where the sinusoidal driver creates a low-rank structure which mutually amplifies the spectral radius until the upper bound imposed by the bounded non-linear activation function is reached. The spectral radius in the numerical simulations approaches 45.455. This suggests that the x_i indeed saturate to approximately ± 1 . The assumption that neural phases align however seems to be weaker, which is likely explained by the non-linear distortion of assumed sinusoidal neural activity.

5.2 EDGE-OF-CHAOS OPTIMALITY FOR SIGNAL RECONSTRUCTION

While we find that Oja learning does not maintain the network near the edge of chaos, we expect that the initial spectral radius g still determines the quality of signal reconstruction during the early learning transient, before the dimension of the weight matrix collapse. For each g_0 , we run the driven Oja system for $T = 80$ time units, ideally long enough for activity transients to decay (since $\tau = 1$) but short enough that Oja has not yet fully saturated the network ($\rho(W) \approx 10\text{--}20$). We then fit a linear readout to reconstruct $y(t) = \sin(\omega t)$ from the activity history $X \in \mathbb{R}^{T/dt \times n}$, the idea being that the set of sinusoidal waves serves as a basis for periodic functions:

$$\mathbf{c}^* = \underset{\mathbf{c}}{\operatorname{argmin}} \|\mathbf{X}\mathbf{c} - \mathbf{y}\|^2, \quad R^2 = 1 - \frac{\|\mathbf{X}\mathbf{c}^* - \mathbf{y}\|^2}{\operatorname{Var}(\mathbf{y})}. \quad (5.2)$$

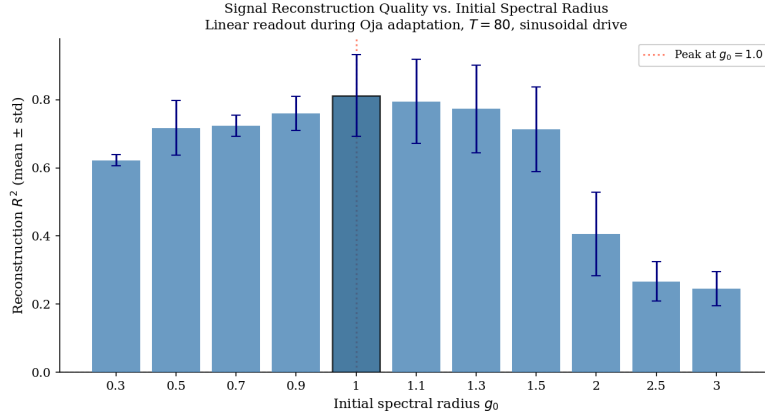


Figure 5.2: Signal reconstruction R^2 as a function of initial spectral radius g_0 .

R^2 peaks at $g_0 \approx 0.9\text{--}1.0$, reaching $R_{\max}^2 \approx 0.81 \pm 0.06$, and falls to $R^2 \approx 0.22$ at $g_0 = 3.0$ (Fig-

ure 5.2). These results suggest that networks initialized at the edge of chaos produce the richest activity during the learning transient, allowing more accurate signal reconstruction before Oja-learning saturates the system, in turn degrading signal diversity. This is consistent with the theoretical reservoir computing model, where proximity to the edge of chaos maximizes the effective useful dimensionality of the network's response **Adi, can you check if this is justified.**

5.3 FROZEN RANDOM RESERVOIR OUTPERFORMS OJA-LEARNED WEIGHTS

To isolate the degradation of the quality in signal reconstruction from Oja-type learning, we compare this with a frozen random reservoir. For each g_0 , we keep the weights fixed (i.e., $\eta, \lambda = 0$) and fit a ridge-regression readout with regularization $\alpha = 10^{-3}$. Ridge-regression was used here to account for potential unstable least-squares regression in the high-dimensional, noisy data.

$$\mathbf{c}^* = (X^T X + \alpha I)^{-1} X^T \mathbf{y}. \quad (5.3)$$

where $\mathbf{y} = \sin(\omega \mathbf{t})$.

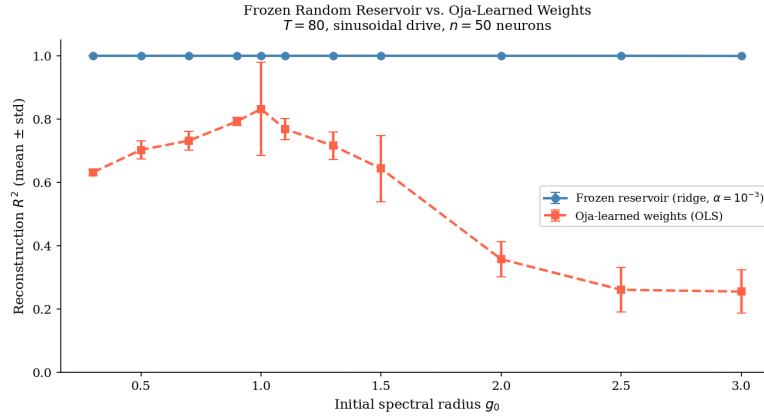


Figure 5.3: Comparison of signal reconstruction R^2 for Oja-learned weights (orange) versus a frozen random reservoir with ridge regression (blue).

The frozen reservoir achieves $R^2 \approx 1.0$ for every g_0 (Figure 5.3). Even this is not a trivial result: the neural network transforms the input sine waves into neural responses with varying amplitude, phases, and nonlinear distortions. On the other hand, Oja-type learning consistently performs worse, driving $\rho(W) \rightarrow 45$, saturating the bounded hyperbolic tangent nonlinearity and degrading the activity diversity in neural activity that allows for high signal reconstruction quality. The frozen network, though, retains useful diversity in neural activity for all g_0 .

5.4 INPUT STRUCTURE DRIVES RANK COLLAPSE

The separation of time scales analysis (Section 4.3) predicts that rank collapse is tied to the rank of $C(W)$, not to the learning rule itself. We test this by replacing the sinusoidal drive with three alternatives:

- *Quasi-periodic*: $\frac{A}{\sqrt{3}} \sum_{k=1}^3 \sin(\omega_k t + \phi_i)$, with $\omega_1 = 1$, $\omega_2 = \varphi$ (golden ratio), $\omega_3 = \sqrt{2}$ (chosen to be mutually incommensurate frequencies).
- *White noise*: $\xi_i(t) \sim \mathcal{N}(0, A^2/2)$ drawn independently for each neuron at each timestep
- *Zero drive*: $A = 0$ (autonomous network).

The parameters of each system here were chosen to match $A \sin(\omega t)$ in power, which we use as a proxy for input strength, in order to isolate the effects of the temporal structure on the system (with obvious exception of the autonomous network).

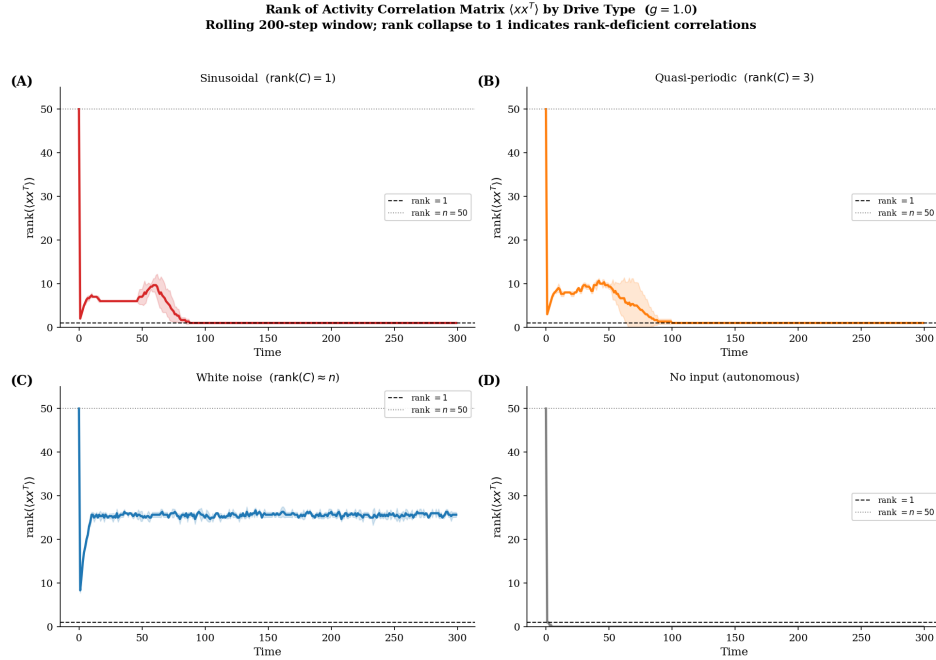


Figure 5.4: Rank of the rolling activity correlation matrix $\hat{C}$ over time for all four drive types at $g = 1.0$, using a 2000-step (100 time unit) rolling window. Sinusoidal (A) and quasi-periodic (B) drives collapse the rank to 1; white noise (C) preserves the full rank $n = 50$; the autonomous network (D) silences to rank 0.

With a 2000-step (100 time unit) rolling window, the results reveal a clear pattern across all four drive types (Figure 5.4). Sinusoidal drive collapses $\text{rank}(\hat{C}) \rightarrow 1$ within $T \approx 100$ time units (panel A). Quasi-periodic drive with three mutually incommensurate frequencies produces

nearly identical collapse, reaching $\text{rank}(\hat{C}) \approx 1$ by $T \approx 150$ time units (panel B). Both structured drives push $\rho(W) \rightarrow 43$ regardless of initial spectral radius.

White noise preserves full-rank correlations: $\text{rank}(\hat{C}) \rightarrow n = 50$ in steady state (panel C). The isotropic structure of white noise gives $C \approx \sigma^2 I$, so $W^* \approx (\eta \sigma^2 / \lambda) I$ is full-rank with bounded spectral radius. In the autonomous case ($A = 0$, panel D), the network silences at $g = 1$: without external drive, the stable fixed point at the origin dominates, yielding $\text{rank}(\hat{C}) \rightarrow 0$.

These results confirm that rank collapse is driven by the temporal structure of the input, not by Oja learning alone. The quasi-periodic result is particularly striking: three mutually incommensurate frequencies are still insufficient to prevent collapse. The crucial distinction is not the number of active spectral modes but whether the input correlation matrix C has low rank—any finite set of periodic components produces a finite-rank C that Oja then amplifies to drive $\rho(W) \rightarrow 45$.

5.5 EFFECTIVE GAIN IS SELF-REGULATED BY SATURATION

In this section, we seek to validate the heuristic effective gain in two complementary experiments.

In the first experiment, we track $g_{\text{eff}}(t)$ during Oja learning alongside $\rho(W(t))$ for sinusoidal and white noise drives. Under sinusoidal drive, we find that $\rho(W)$ approaches approximately 43 but every neuron saturates ($|h_i| \rightarrow \infty$), so $d_i = \text{sech}^2(h_i) \rightarrow 0$ and $g_{\text{eff}} \rightarrow 0$. Here, the system is in a maximally saturated state where the linearized perturbation dynamics are always stable despite the exploding nominal spectral radius.

In our second experiment, we freeze the weights ($\eta, \lambda = 0$) and scan $g \in [0.1, 3.0]$ with sinusoidal drive, averaging g_{eff} and $\rho(D(t)W)$ over the steady-state window $t \in [210, 300]$. **Quickly revise the first experiment to describe if the same thing is done**

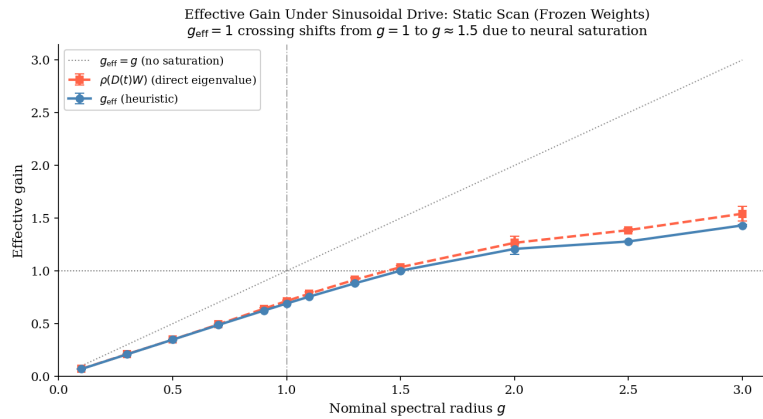


Figure 5.5: Time-averaged effective gain g_{eff} (blue) and directly calculated spectral radius $\rho(D(t)W)$ (orange) versus nominal g for frozen weights under sinusoidal drive. The dotted diagonal shows the no-saturation baseline $g_{\text{eff}} = g$.

Here we see that heuristic tracks the "effective" spectral radius of the system, $\rho(D(t)W(t))$, within 5% across all tested g (Figure 5.5). The $g_{\text{eff}} = 1$ crossing occurs at $g \approx 1.5$: sinusoidal

drive saturates neurons enough to reduce the effective gain by approximately 33%, shifting the effective stability boundary substantially above the Girko prediction $g = 1$. **How is g calculated here?** **Revise the first section slightly to clarify that weights here were frozen to isolate the DW dynamics**

5.6 CRITICAL SLOWING DOWN AT THE PHASE TRANSITION

Independent evidence for the $g = 1$ phase transition comes from the autonomous discrete map $h_{t+1} = Wf(h_t)$. The largest Lyapunov exponent (LLE) λ_1 is estimated via a 300-step warmup followed by 800-step single-vector power iteration on the tangent map $J_t = W \text{diag}(\text{sech}^2(h_t))$. For $\lambda_1 < 0$ the system is stable with relaxation time $\tau_{\text{relax}} = -1/\lambda_1$; for $\lambda_1 > 0$ the map is chaotic. In the linear regime a small perturbation $\delta h(0)$ is predicted to grow as $\|\delta h(t)\| \approx \varepsilon e^{\lambda_1 t}$.

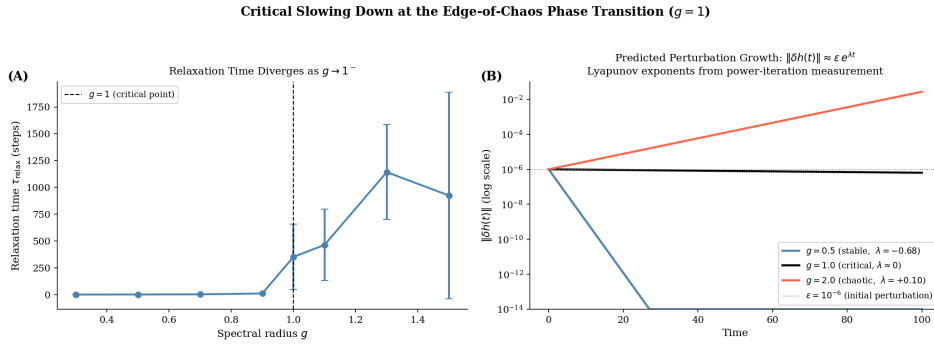


Figure 5.6: Left: relaxation time $\tau_{\text{relax}} = -1/\lambda_1$ versus g for the autonomous discrete map; τ_{relax} diverges as $g \rightarrow 1^-$. Right: predicted perturbation growth $\|\delta h(t)\| = \varepsilon e^{\lambda_1 t}$ using the power-iteration LLE values, illustrating the stable ($g = 0.5$), critical ($g = 1.0$), and chaotic ($g = 2.0$) regimes.

Table 5.1: Largest Lyapunov exponent and relaxation time for selected g values ($n = 50$, autonomous discrete map, mean over 5 seeds).

g	λ_1 (nats/step)	τ_{relax} (steps)
0.30	-1.20	0.8
0.70	-0.35	2.9
0.90	-0.10	10.8
1.00	-0.005	≈ 350
2.00	+0.10	(chaotic)

Table 5.1 and Figure 5.6 show that τ_{relax} diverges as $g \rightarrow 1^-$, which is the defining signature of a continuous (second-order) phase transition [?]. At $g = 1$ the network has no preferred timescale for returning to equilibrium: small disturbances persist arbitrarily long. This criti-

cal slowing down provides a physically clean signature of the edge-of-chaos transition that is observable independently of the sign of the LLE.

A note on finite-size effects: Sompolinsky’s mean-field theory [?] predicts chaos onset exactly at $g = 1$ in the $n \rightarrow \infty$ limit. For $n = 50$, the discrete map remains stable up to $g \approx 1.5$ (LLE ≈ -0.009); positive LLE first appears near $g \approx 2.0$ (Table 5.1). The τ_{relax} divergence near $g = 1$ is nonetheless clearly visible and confirms the qualitative picture even at finite n .

6 CONCLUSIONS